Return-to-Play After Concussion

```r
library(tidyverse)
library(brms)
library(tidybayes)
library(marginaleffects)
```

Data

One row per concussion event. Athletes with no concussions are excluded — they contribute no outcome to the model.

```r
rtp_data <- read_csv("data/concussion_data.csv", show_col_types = FALSE) |>
  filter(!is.na(injury_date)) |>
  mutate(
    sex  = factor(sex, levels = c("M", "F")),
    city = factor(city)
  )

glimpse(rtp_data)
```

```
Rows: 63
Columns: 13
$ athlete_id          <chr> "CWG018", "SZE093", "BEV650", "GRY627", "MSW762", …
$ sex                 <fct> M, F, F, F, F, F, F, F, M, M, M, F, F, F, F, F, M,…
$ height_cm           <dbl> 174, 176, 168, 162, 179, 169, 169, 169, 193, 193, …
$ birthdate           <dttm> 2005-06-05 10:04:17, 2006-11-30 10:04:17, 2002-06…
$ city                <fct> Boston, Atlanta, Boston, Denver, San Diego, San Di…
$ sport               <chr> "Lacrosse", "Soccer", "Lacrosse", "Flag Football",…
$ total_games         <dbl> 36, 45, 66, 31, 98, 101, 101, 101, 96, 96, 44, 43,…
$ concussion_num      <dbl> 1, 1, 1, 1, 1, 1, 2, 3, 1, 2, 1, 1, 1, 1, 2, 1, 1,…
$ injury_date         <date> 2023-02-17, 2024-07-30, 2022-06-25, 2011-09-01, 2…
$ return_to_play_date <date> 2023-03-01, 2024-08-24, 2022-07-23, 2011-09-05, 2…
$ days_to_rtp         <dbl> 12, 25, 28, 4, 8, 15, 22, 17, 6, 13, 4, 10, 18, 7,…
$ scat5_initial       <dbl> 92, 51, 63, 19, 28, 48, 47, 54, 79, 62, 22, 38, 53…
$ scat5_return        <dbl> 12, 6, 18, 5, 8, 9, 5, 7, 9, 8, 4, 8, 9, 5, 5, 13,…
```

Model

Lognormal regression for `days_to_rtp`. On the log scale, effects are additive and exponentiate to multiplicative shifts in median RTP days.

Fixed effects:

- `scat5_initial` — injury severity drives longer recovery
- `sex` — female athletes recover more slowly at equivalent severity
- `concussion_num` — each repeat concussion extends recovery

Random effect:

- `(1 | city)` — training center captures unexplained protocol variation across the 8 centers

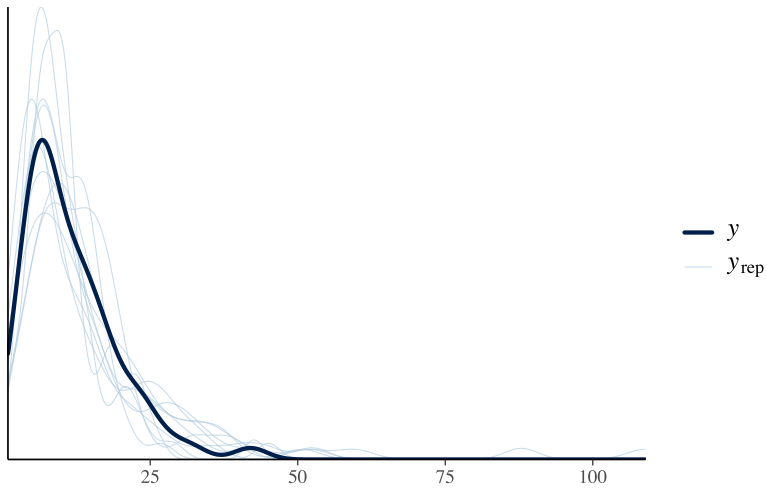
```
#|
fit <- brm(
  days_to_rtp ~ scat5_initial + sex + concussion_num + (1 | city),
  family = lognormal(),
  data = rtp_data,
  chains = 4,
  cores = 4,
  backend = "cmdstan"
)
```

```
summary(fit)
```

```
 Family: lognormal
  Links: mu = identity
Formula: days_to_rtp ~ scat5_initial + sex + concussion_num + (1 | city)
   Data: rtp_data (Number of observations: 63)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~city (Number of levels: 8)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.30      0.17     0.04     0.73 1.00      838      885

Regression Coefficients:
               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept          1.03      0.32     0.40     1.67 1.00     3299     2555
scat5_initial      0.02      0.00     0.01     0.02 1.00     4741     3058
sexF               0.27      0.15    -0.03     0.57 1.00     3759     2715
concussion_num     0.26      0.16    -0.06     0.58 1.00     4203     2736

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.57      0.06     0.47     0.70 1.00     3088     2823
```

```
Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
pp_check(fit)
```



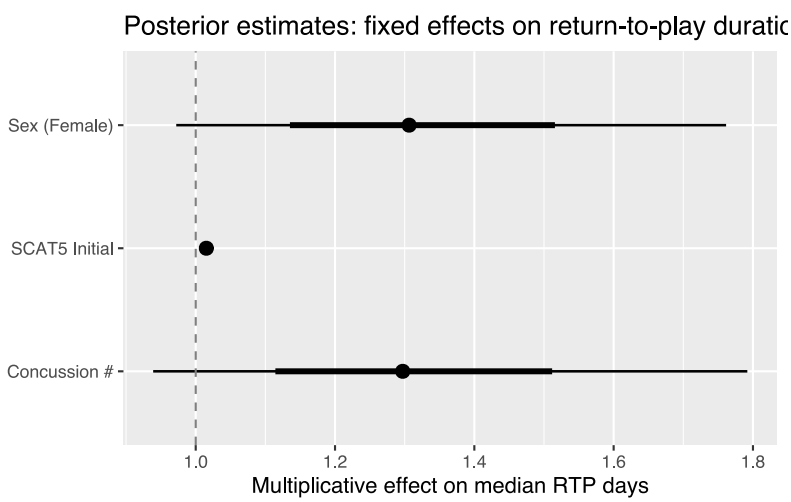
Fixed Effects

Exponentiated coefficients are multiplicative factors on median RTP days. A coefficient of 1.22 means 22% longer recovery.

```
fit |>
  as_draws_df() |>
  select(b_Intercept, b_scat5_initial, b_sexF, b_concussion_num) |>
  pivot_longer(everything()) |>
  mutate(value = exp(value)) |>
  summarise(
    median = median(value),
    q2.5   = quantile(value, 0.025),
    q97.5  = quantile(value, 0.975),
    .by = name
  ) |>
  mutate(across(where(is.numeric), \(x) round(x, 3))) |>
  arrange(desc(median)) |>
  knitr::kable()
```

name	median	q2.5	q97.5
b_Intercept	2.787	1.495	5.336
b_sexF	1.306	0.972	1.761
b_concussion_num	1.297	0.939	1.792
b_scat5_initial	1.015	1.007	1.024

```
fit |>
  as_draws_df() |>
  select(b_scat5_initial, b_sexF, b_concussion_num) |>
  pivot_longer(everything()) |>
  mutate(name = case_match(
    name,
    "b_scat5_initial"  ~ "SCAT5 Initial",
    "b_sexF"           ~ "Sex (Female)",
    "b_concussion_num" ~ "Concussion #"
  )) |>
  ggplot(aes(exp(value), name)) +
  stat_pointinterval() +
  geom_vline(xintercept = 1, linetype = "dashed", color = "grey50") +
  labs(
    x = "Multiplicative effect on median RTP days",
    y = NULL,
    title = "Posterior estimates: fixed effects on return-to-play duration"
  )
```



Sex Effect

Marginal effect of sex on predicted RTP days, averaging over the observed distribution of severity and concussion history.

```
avg_comparisons(fit, variables = "sex") |>
  tibble() |>
  select(contrast, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 1))) |>
  knitr::kable()
```

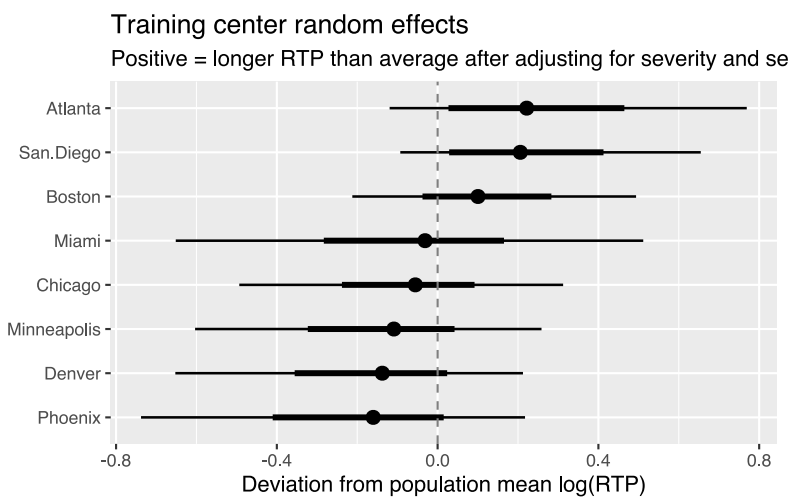
contrast	estimate	conf.low	conf.high
F - M	3.2	−0.3	7.3

Training Center Random Effects

Posterior estimates of each center's deviation from the population mean RTP, on the log scale. Positive values indicate longer-than-average RTP at that center after accounting for athlete severity and sex.

```
fit |>
  spread_draws(r_city[city, ]) |>
  ungroup() |>
  summarise(
    median = median(r_city),
    q2.5   = quantile(r_city, 0.025),
    q97.5  = quantile(r_city, 0.975),
    .by = city
  ) |>
  arrange(desc(median)) |>
  mutate(across(where(is.numeric), \(x) round(x, 3))) |>
  knitr::kable()
```

city	median	q2.5	q97.5
Atlanta	0.222	−0.119	0.769
San.Diego	0.206	−0.093	0.655
Boston	0.100	−0.212	0.494
Miami	−0.031	−0.652	0.512
Chicago	−0.055	−0.493	0.312
Minneapolis	−0.109	−0.603	0.258
Denver	−0.138	−0.653	0.212

city	median	q2.5	q97.5
Phoenix	−0.160	−0.738	0.217

```
fit |>
  spread_draws(r_city[city, ]) |>
  ggplot(aes(r_city, reorder(city, r_city))) +
  stat_pointinterval() +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey50") +
  labs(
    x = "Deviation from population mean log(RTP)",
    y = NULL,
    title = "Training center random effects",
    subtitle = "Positive = longer RTP than average after adjusting for severity
and sex"
  )
```